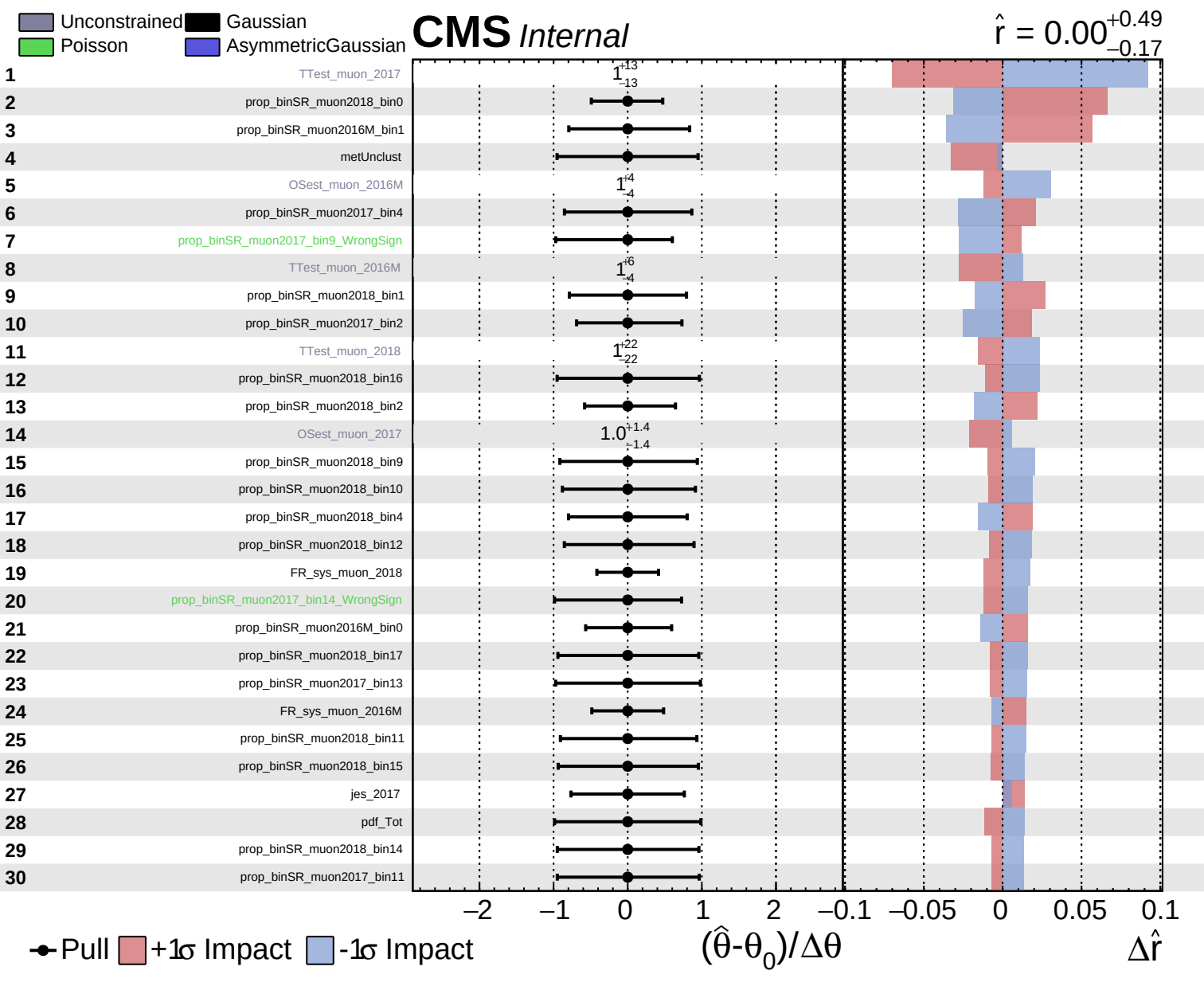
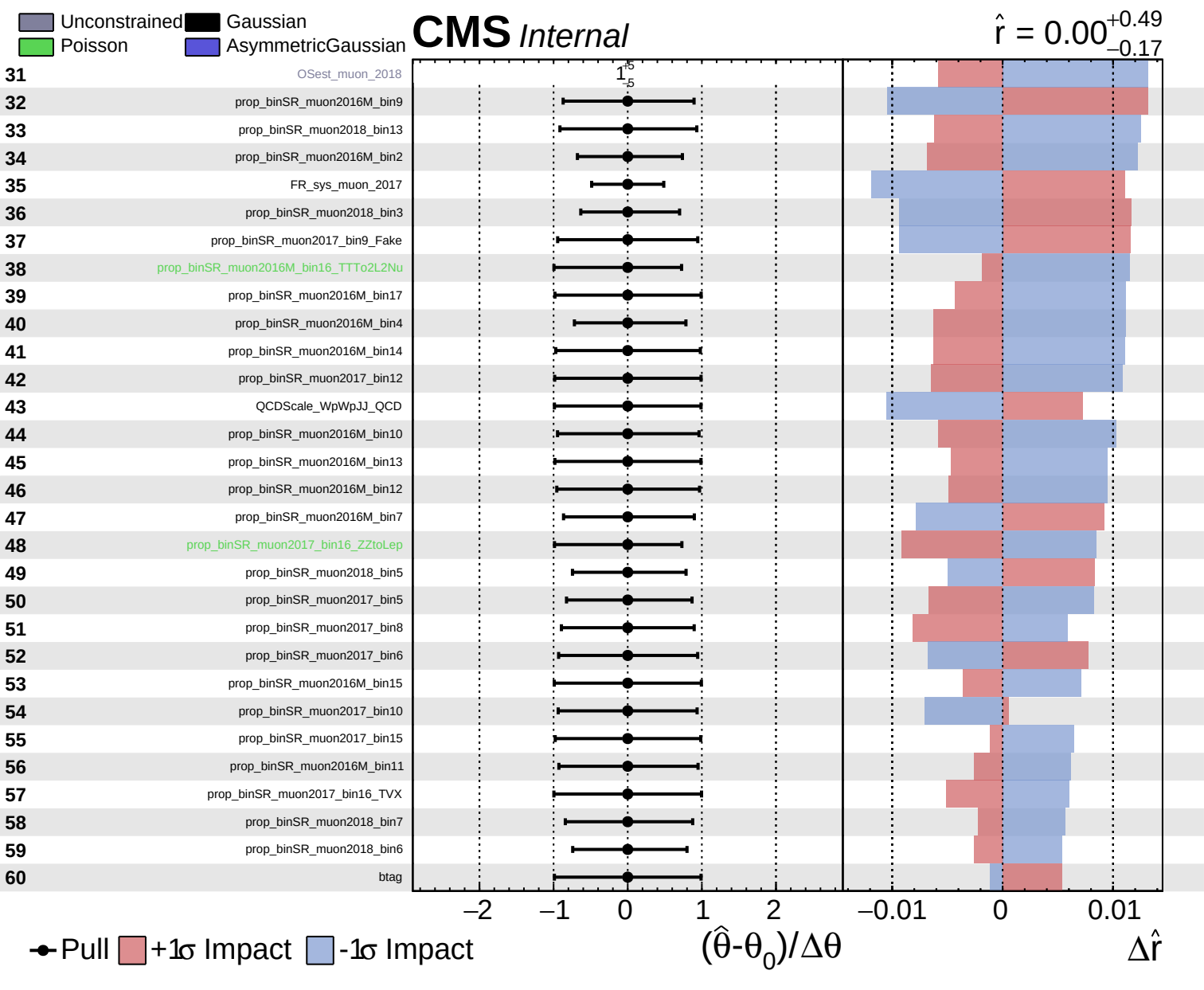
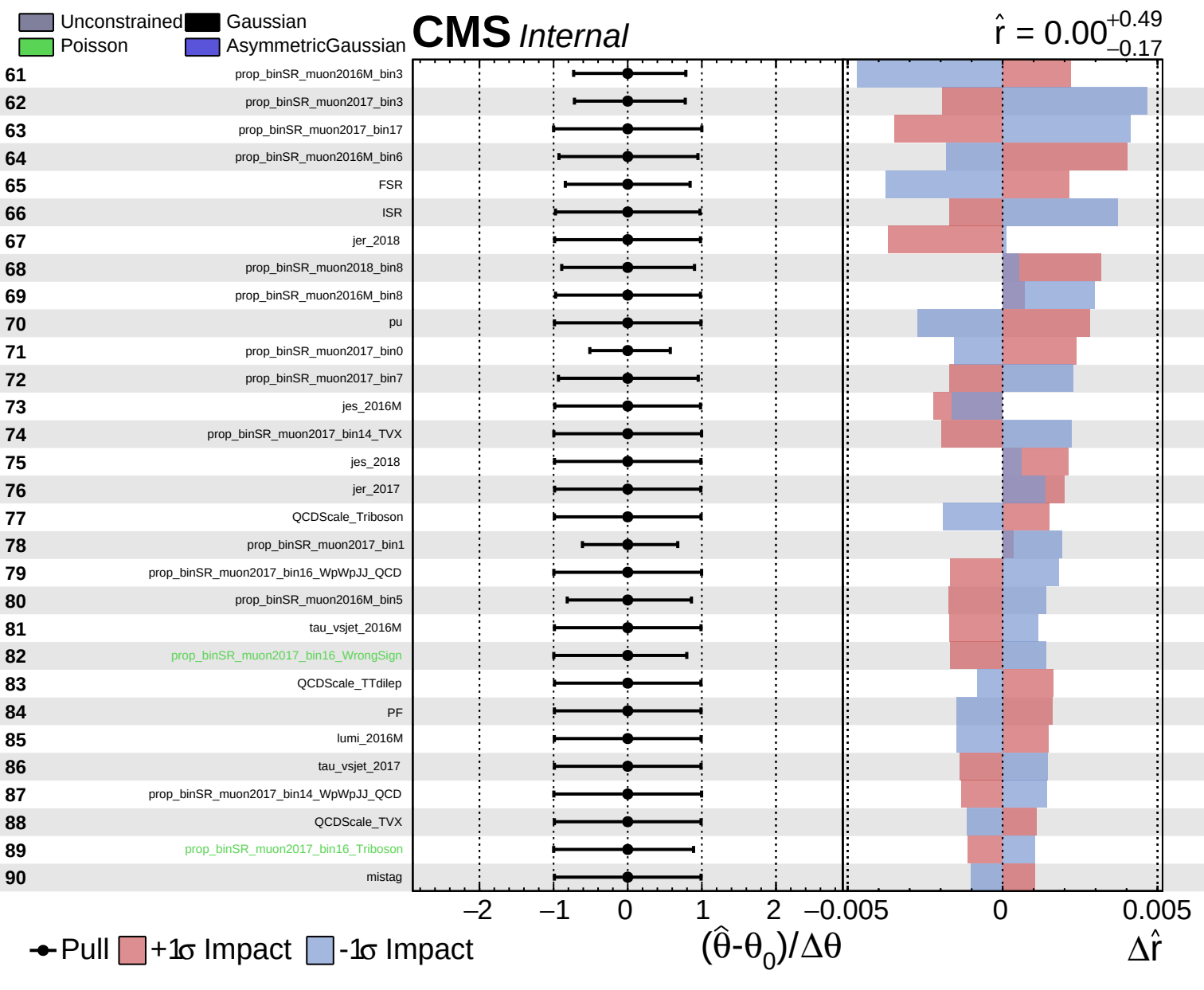
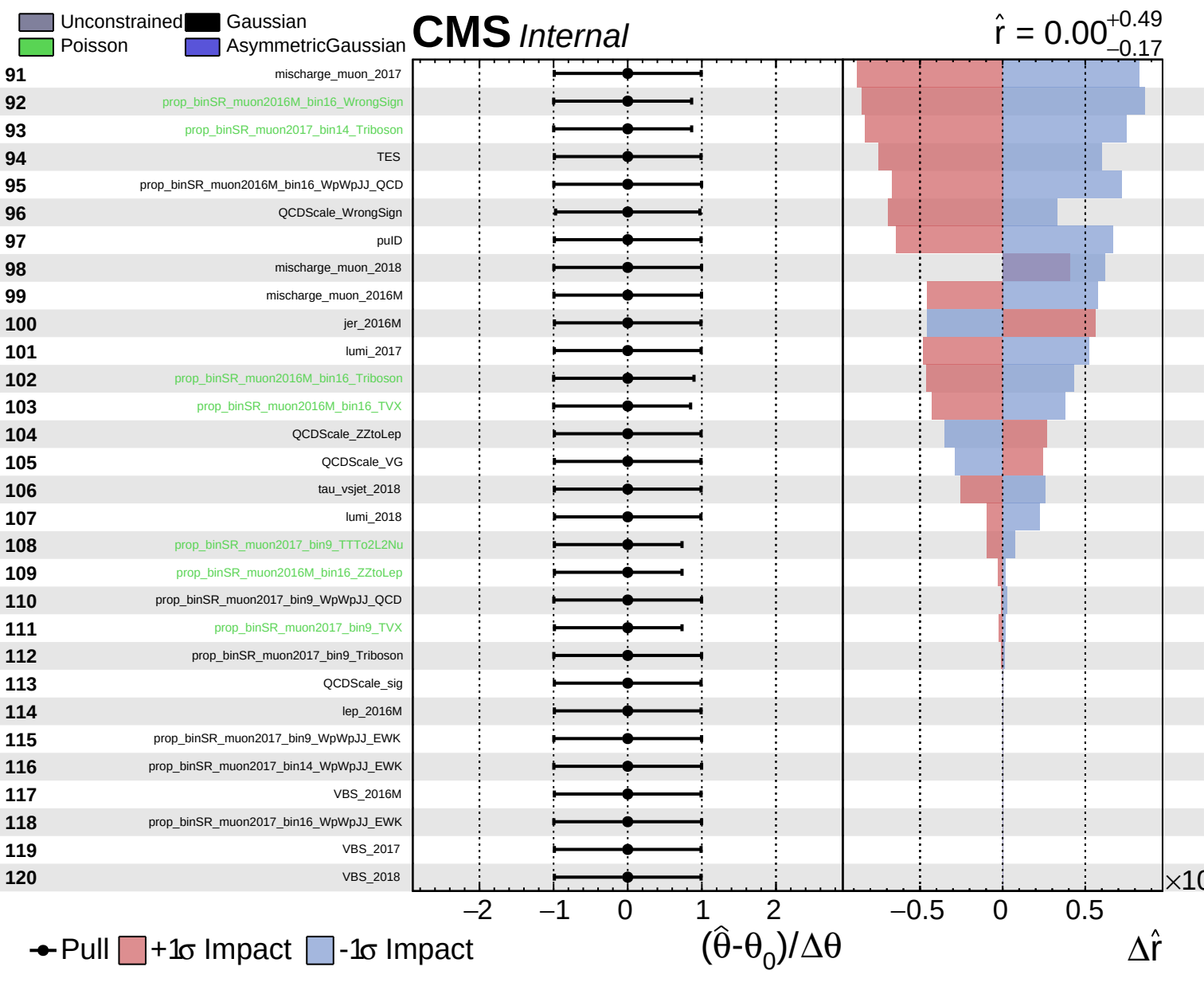
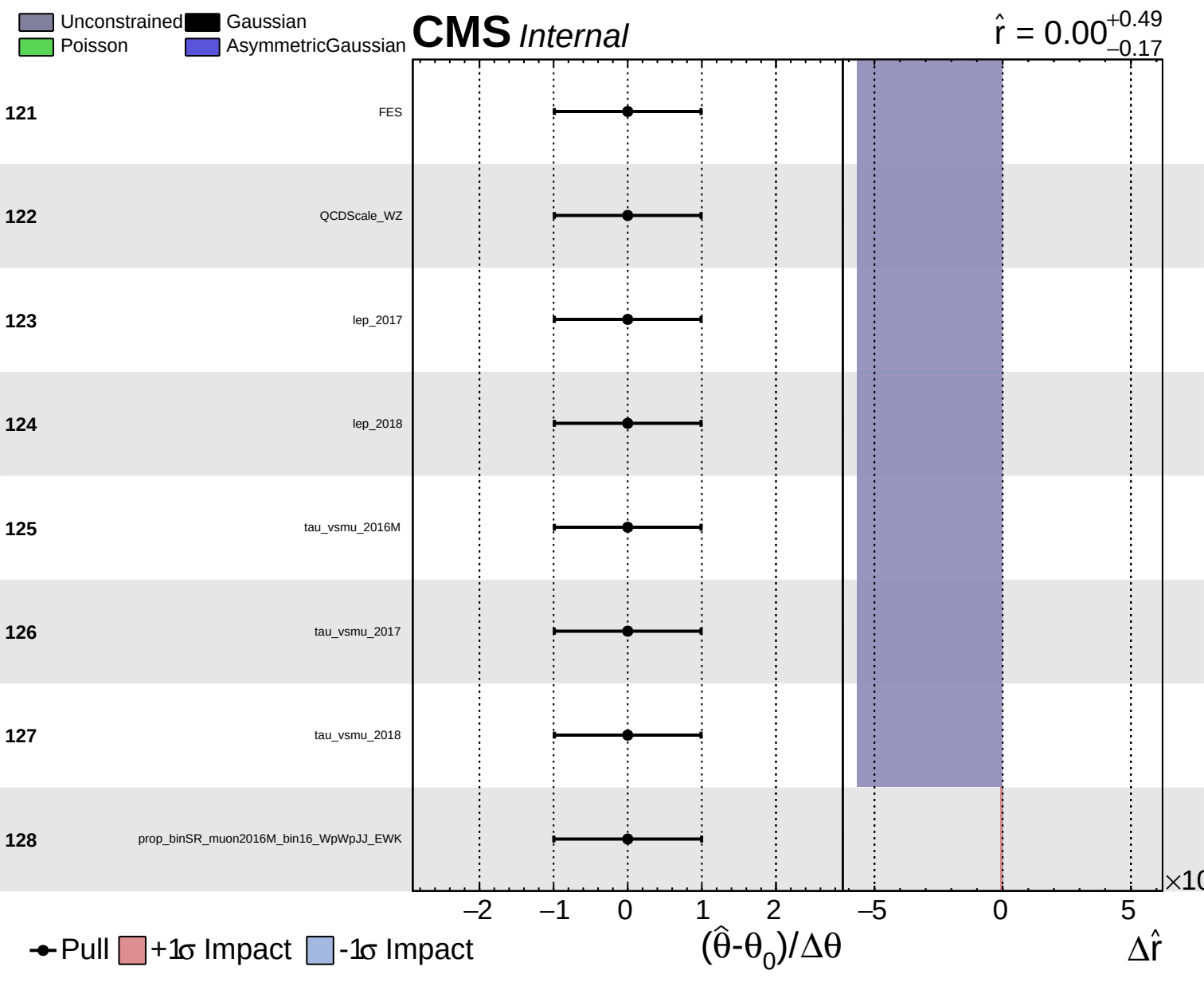

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.49 −0.17
1 TTest_muon_2017
2 prop_binSR_muon2018_bin0
3 prop_binSR_muon2016M_bin1
4 metUnclust
5 OSest_muon_2016M
6 prop_binSR_muon2017_bin4
7 prop_binSR_muon2017_bin9_WrongSign
8 TTest_muon_2016M
9 prop_binSR_muon2018_bin1
10 prop_binSR_muon2017_bin2
11 TTest_muon_2018
12 prop_binSR_muon2018_bin16
13 prop_binSR_muon2018_bin2
14 OSest_muon_2017
15 prop_binSR_muon2018_bin9
16 prop_binSR_muon2018_bin10
17 prop_binSR_muon2018_bin4
18 prop_binSR_muon2018_bin12
19 FR_sys_muon_2018
20 prop_binSR_muon2017_bin14_WrongSign
21 prop_binSR_muon2016M_bin0
22 prop_binSR_muon2018_bin17
23 prop_binSR_muon2017_bin13
24 FR_sys_muon_2016M
25 prop_binSR_muon2018_bin11
26 prop_binSR_muon2018_bin15
27 jes_2017
28 pdf_Tot
29 prop_binSR_muon2018_bin14
30 prop_binSR_muon2017_bin11
1 +13 −13
1 +4 −4
1 +6 −4
1 +22 −22
1.0 +1.4 −1.4
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.49 −0.17
31 OSest_muon_2018
32 prop_binSR_muon2016M_bin9
33 prop_binSR_muon2018_bin13
34 prop_binSR_muon2016M_bin2
35 FR_sys_muon_2017
36 prop_binSR_muon2018_bin3
37 prop_binSR_muon2017_bin9_Fake
38 prop_binSR_muon2016M_bin16_TTTo2L2Nu
39 prop_binSR_muon2016M_bin17
40 prop_binSR_muon2016M_bin4
41 prop_binSR_muon2016M_bin14
42 prop_binSR_muon2017_bin12
43 QCDScale_WpWpJJ_QCD
44 prop_binSR_muon2016M_bin10
45 prop_binSR_muon2016M_bin13
46 prop_binSR_muon2016M_bin12
47 prop_binSR_muon2016M_bin7
48 prop_binSR_muon2017_bin16_ZZtoLep
49 prop_binSR_muon2018_bin5
50 prop_binSR_muon2017_bin5
51 prop_binSR_muon2017_bin8
52 prop_binSR_muon2017_bin6
53 prop_binSR_muon2016M_bin15
54 prop_binSR_muon2017_bin10
55 prop_binSR_muon2017_bin15
56 prop_binSR_muon2016M_bin11
57 prop_binSR_muon2017_bin16_TVX
58 prop_binSR_muon2018_bin7
59 prop_binSR_muon2018_bin6
60 btag
1 +5 −5
−2 −1 0 1 2
−0.01 0 0.01
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δr̂


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.49 −0.17
61 prop_binSR_muon2016M_bin3
62 prop_binSR_muon2017_bin3
63 prop_binSR_muon2017_bin17
64 prop_binSR_muon2016M_bin6
65 FSR
66 ISR
67 jer_2018
68 prop_binSR_muon2018_bin8
69 prop_binSR_muon2016M_bin8
70 pu
71 prop_binSR_muon2017_bin0
72 prop_binSR_muon2017_bin7
73 jes_2016M
74 prop_binSR_muon2017_bin14_TVX
75 jes_2018
76 jer_2017
77 QCDScale_Triboson
78 prop_binSR_muon2017_bin1
79 prop_binSR_muon2017_bin16_WpWpJJ_QCD
80 prop_binSR_muon2016M_bin5
81 tau_vsjet_2016M
82 prop_binSR_muon2017_bin16_WrongSign
83 QCDScale_TTdilep
84 PF
85 lumi_2016M
86 tau_vsjet_2017
87 prop_binSR_muon2017_bin14_WpWpJJ_QCD
88 QCDScale_TVX
89 prop_binSR_muon2017_bin16_Triboson
90 mistag
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.49 −0.17
91 mischarge_muon_2017
92 prop_binSR_muon2016M_bin16_WrongSign
93 prop_binSR_muon2017_bin14_Triboson
94 TES
95 prop_binSR_muon2016M_bin16_WpWpJJ_QCD
96 QCDScale_WrongSign
97 puID
98 mischarge_muon_2018
99 mischarge_muon_2016M
100 jer_2016M
101 lumi_2017
102 prop_binSR_muon2016M_bin16_Triboson
103 prop_binSR_muon2016M_bin16_TVX
104 QCDScale_ZZtoLep
105 QCDScale_VG
106 tau_vsjet_2018
107 lumi_2018
108 prop_binSR_muon2017_bin9_TTTo2L2Nu
109 prop_binSR_muon2016M_bin16_ZZtoLep
110 prop_binSR_muon2017_bin9_WpWpJJ_QCD
111 prop_binSR_muon2017_bin9_TVX
112 prop_binSR_muon2017_bin9_Triboson
113 QCDScale_sig
114 lep_2016M
115 prop_binSR_muon2017_bin9_WpWpJJ_EWK
116 prop_binSR_muon2017_bin14_WpWpJJ_EWK
117 VBS_2016M
118 prop_binSR_muon2017_bin16_WpWpJJ_EWK
119 VBS_2017
120 VBS_2018
×10
−2 −1 0 1 2
−0.5 0 0.5
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δr̂


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.49 −0.17
121 FES
122 QCDScale_WZ
123 lep_2017
124 lep_2018
125 tau_vsmu_2016M
126 tau_vsmu_2017
127 tau_vsmu_2018
128 prop_binSR_muon2016M_bin16_WpWpJJ_EWK
×10
−2 −1 0 1 2
−5 0 5
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δr̂